NexLearn LMS

Class Model Document

UE23CS352B — Object Oriented Analysis & Design

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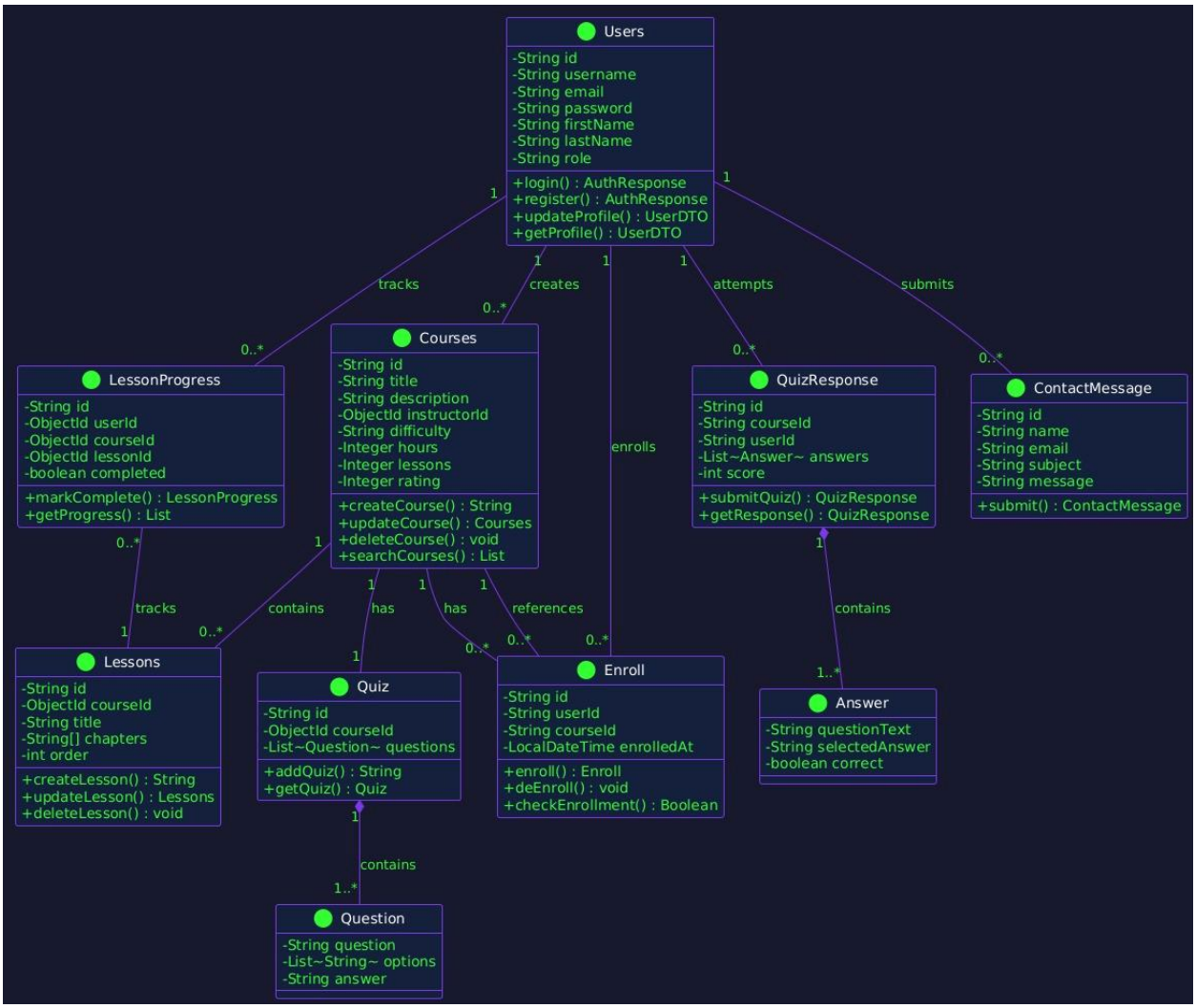
Technology: Spring Boot 3.4 + Next.js 15 + MongoDB

Class Diagram



Class Model Specifications

This document provides detailed class model specifications for all 10 domain classes in the NexLearn Learning Management System. Each class corresponds to a MongoDB document collection (or embedded object) in the backend, implemented using Spring Boot 3.4 with Spring Data MongoDB.

1. Users

Stereotype: Entity / Domain Class
Layer: Model (MongoDB Document — collection: users)

Description

Represents both Student and Instructor accounts. Stores authentication credentials and role information. Central entity in the system — all use cases trace back to a User.

Attributes

Visibility	Attribute	Type	Description
–	`id`	String	Auto-generated MongoDB ObjectId
–	`username`	String	Unique login identifier
–	`email`	String	Unique email address
–	`password`	String	Plain text password (demo environment)
–	`firstName`	String	User's first name
–	`lastName`	String	User's last name
–	`role`	String	STUDENT or INSTRUCTOR

Methods

Visibility	Method	Return Type	Description
+	`login()`	AuthResponse	Validates credentials, generates and returns JWT token
+	`register()`	AuthResponse	Creates new account, returns JWT token
+	`updateProfile()`	UserDTO	Updates firstName, lastName, or password
+	`getProfile()`	UserDTO	Returns public user data (no password)

Relationships

- 1 User creates 0..* Courses (as Instructor)
- 1 User tracks 0..* LessonProgress records (as Student)
- 1 User attempts 0..* QuizResponses (as Student)
- 1 User submits 0..* ContactMessages
- 1 User enrolls in 0..* Courses via Enroll association class

2. Courses

Stereotype: Entity / Domain Class
Layer: Model (MongoDB Document — collection: courses)

Description

Represents a published course in the system. Owned by an Instructor and consumed by Students. Central to UC-03 (creation), UC-04 (enrollment), UC-05 (content), UC-07 (monitoring), and UC-08 (search).

Attributes

Visibility	Attribute	Type	Description
–	`id`	String	Auto-generated MongoDB ObjectId
–	`title`	String	Course title (min 5 chars)
–	`description`	String	Course description (min 20 chars)
–	`instructorId`	ObjectId	Foreign key reference to Users
–	`difficulty`	String	Beginner / Intermediate / Advanced
–	`hours`	Integer	Total estimated course hours
–	`lessons`	Integer	Total lesson count
–	`rating`	Integer	Course rating out of 5

Methods

Visibility	Method	Return Type	Description
+	`createCourse()`	String	Persists new course, returns generated id
+	`updateCourse()`	Courses	Updates course fields, returns updated object
+	`deleteCourse()`	void	Removes course document from MongoDB
+	`searchCourses()`	List	Regex search on title and description fields

Relationships

- 1 Course contains 1..* Lessons (composition)
- 1 Course has exactly 1 Quiz (composition)
- 1 Course has 0..* Enroll records (association)
- 1 Course is referenced by 0..* LessonProgress records

3. Lessons

Stereotype: Entity / Domain Class
Layer: Model (MongoDB Document — collection: lessons)

Description

Represents a single lesson within a course. Contains an ordered array of chapter names. Students check off chapters and mark lessons complete. Supports UC-05 (content upload) and UC-06 (progress tracking).

Attributes

Visibility	Attribute	Type	Description
–	`id`	String	Auto-generated MongoDB ObjectId
–	`courseId`	ObjectId	Foreign key reference to Courses
–	`title`	String	Lesson title
–	`chapters`	String[]	Ordered array of chapter names
–	`order`	int	Display order within parent course

Methods

Visibility	Method	Return Type	Description
+	`createLesson()`	String	Persists lesson, returns generated id
+	`updateLesson()`	Lessons	Updates lesson title or chapters
+	`deleteLesson()`	void	Removes lesson document from MongoDB

Relationships

- 0..* Lessons belong to 1 Course
- 0..* LessonProgress records track 1 Lesson

4. Enroll

Stereotype: Association Class
Layer: Model (MongoDB Document — collection: enroll)

Description

Reifies the many-to-many enrollment relationship between Users and Courses. Stores timestamp of enrollment. Supports UC-04 (enroll), UC-06 (dashboard), and UC-07 (student monitoring).

Attributes

Visibility	Attribute	Type	Description
–	`id`	String	Auto-generated MongoDB ObjectId
–	`userId`	String	Foreign key reference to Users
–	`courseId`	String	Foreign key reference to Courses
–	`enrolledAt`	LocalDateTime	Timestamp when enrollment was created

Methods

Visibility	Method	Return Type	Description
+	`enroll()`	Enroll	Creates enrollment record, prevents duplicates (409)
+	`deEnroll()`	void	Deletes enrollment (only allowed if no progress exists)
+	`checkEnrollment()`	Boolean	Returns true if userId+courseId pair exists

Relationships

- 0..* Enroll records reference 1 Course
- 0..* Enroll records belong to 1 User

5. LessonProgress

Stereotype: Entity / Domain Class
Layer: Model (MongoDB Document — collection: lesson_progress)

Description

Tracks whether a specific Student has completed a specific Lesson within a Course. The lessonId is serialized as a hex string for consistent frontend map key usage. Supports UC-06 (progress tracking) and UC-07 (student monitoring).

Attributes

Visibility	Attribute	Type	Description
–	`id`	String	Auto-generated MongoDB ObjectId (String)
–	`userId`	ObjectId	Foreign key reference to Users
–	`courseId`	ObjectId	Foreign key reference to Courses
–	`lessonId`	ObjectId	Foreign key reference to Lessons
–	`completed`	boolean	True when lesson is marked complete

Methods

Visibility	Method	Return Type	Description
+	`markComplete()`	LessonProgress	Sets completed=true, upserts record
+	`getProgress()`	List	Returns all LessonProgress for userId+courseId

Relationships

- 0..* LessonProgress records track 1 Lesson
- 0..* LessonProgress records belong to 1 User

6. Quiz

Stereotype: Entity / Domain Class
Layer: Model (MongoDB Document — collection: quiz)

Description

Represents the end-of-course quiz for a specific Course. Contains an embedded list of Question objects. Only accessible to students after all lessons are marked complete. Supports UC-03 (quiz creation) and UC-06 (quiz taking).

Attributes

Visibility	Attribute	Type	Description
–	`id`	String	Auto-generated MongoDB ObjectId
–	`courseId`	ObjectId	Foreign key reference to Courses
–	`questions`	List<Question>	Embedded list of Question objects

Methods

Visibility	Method	Return Type	Description
+	`addQuiz()`	String	Persists quiz document, returns generated id
+	`getQuiz()`	Quiz	Retrieves quiz by courseId

Relationships

- 1 Quiz belongs to exactly 1 Course
- 1 Quiz contains 1..* Questions (composition — embedded documents)

7. Question

Stereotype: Value Object / Embedded Class
Layer: Embedded inside Quiz document (no independent collection)

Description

Represents a single quiz question with four answer options and one correct answer. Embedded directly inside Quiz — has no independent identity or MongoDB collection. Part of the composition relationship with Quiz.

Attributes

Visibility	Attribute	Type	Description
–	`question`	String	The question text
–	`options`	List<String>	Exactly four answer choices
–	`answer`	String	The correct answer text

Relationships

- 1..* Questions are contained within 1 Quiz (composition, embedded document)
- No independent existence — deleted when parent Quiz is deleted

8. QuizResponse

Stereotype: Entity / Domain Class
Layer: Model (MongoDB Document — collection: quiz_responses)

Description

Records a Student's quiz submission for a specific Course. Stores all answers and the calculated score. Supports upsert — retaking a quiz overwrites the previous response. Supports UC-06 (quiz completion).

Attributes

Visibility	Attribute	Type	Description
–	`id`	String	Auto-generated MongoDB ObjectId
–	`courseId`	String	Foreign key reference to Courses
–	`userId`	String	Foreign key reference to Users
–	`answers`	List<Answer>	Embedded list of Answer objects
–	`score`	int	Count of correct answers

Methods

Visibility	Method	Return Type	Description
+	`submitQuiz()`	QuizResponse	Scores answers and upserts response document
+	`getResponse()`	QuizResponse	Retrieves response by courseId + userId

Relationships

- 0..* QuizResponses belong to 1 User
- 1 QuizResponse contains 1..* Answers (composition — embedded documents)

9. Answer

Stereotype: Value Object / Embedded Class
Layer: Embedded inside QuizResponse document (no independent collection)

Description

Records a Student's response to a single Question. Embedded directly inside QuizResponse. The correct field is set by QuizService when comparing selectedAnswer against the Quiz's question answer. Part of the composition with QuizResponse.

Attributes

Visibility	Attribute	Type	Description
–	`questionText`	String	The question that was answered
–	`selectedAnswer`	String	The student's chosen answer
–	`correct`	boolean	Whether the selected answer matches the correct answer

Relationships

- 1..* Answers are contained within 1 QuizResponse (composition, embedded document)
- No independent existence — deleted when parent QuizResponse is deleted

10. ContactMessage

Stereotype: Entity / Domain Class
Layer: Model (MongoDB Document — collection: contact_messages)

Description

Records a support or feedback message submitted via the Contact page. Accessible without authentication (pre-conditions from UC-10 spec). Persisted to MongoDB for follow-up. Supports UC-10 (Contact and Support).

Attributes

Visibility	Attribute	Type	Description
–	`id`	String	Auto-generated MongoDB ObjectId
–	`name`	String	Sender's full name
–	`email`	String	Sender's email address
–	`subject`	String	Message subject line
–	`message`	String	Full message body

Methods

Visibility	Method	Return Type	Description
+	`submit()`	ContactMessage	Validates fields and persists message to MongoDB

Relationships

- 0..* ContactMessages are submitted by 1 User (or anonymous visitor)

Relationship Summary

From	To	Type	Multiplicity	Label
Users	`Courses`	Association	1 → 0..*	creates
Users	`LessonProgress`	Association	1 → 0..*	tracks
Users	`QuizResponse`	Association	1 → 0..*	attempts
Users	`ContactMessage`	Association	1 → 0..*	submits
Users	`Enroll`	Association	1 → 0..*	enrolls
Courses	`Lessons`	Composition	1 → 1..*	contains
Courses	`Quiz`	Composition	1 → 1	has
Courses	`Enroll`	Association	1 → 0..*	has
Courses	`LessonProgress`	Association	1 → 0..*	referenced by
Quiz	`Question`	Composition	1 → 1..*	contains
QuizResponse	`Answer`	Composition	1 → 1..*	contains
Enroll	`Courses`	Association	0..* → 1	references
LessonProgress	`Lessons`	Association	0..* → 1	tracks